

Darsh Kapadia

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EDUCATION

Illinois Institute of Technology, MS, Cyber Forensics & Security; GPA: 4.00 Jan 2025 – May 2027
Illinois Institute of Technology, BS, Applied Cyber Security & IT; GPA: 3.96 Aug 2023 – May 2027
Selected Coursework: Active Cyber Defense, Programming for Cyber Analytics, Cyber Forensics, Vulnerability Analysis and Control, Coding Security, Digital Evidence, AI for Cybersecurity;
Current Coursework: Information Technology Auditing, Operating System Security, Database Security, Zero Trust Architecture

EXPERIENCE

Cyber Security Intern, Capital One – McLean, VA Jun 2026 – Aug 2026

- Pioneered a detection-health evaluation methodology that made detection fidelity actionable for engineers and leadership by building and validating a Databricks dashboard in Python and Apache Spark that modeled bursty and data-starved alert volumes with a hybrid statistical model.
- Operationalized the methodology with Confluence documentation, an interactive HTML explainer, a probability-and-statistics whitepaper, demonstration videos, and a Claude skill for notebook workflows to support consistent understanding and reuse by engineers and leadership.
- Provided the team with a full year of lead time to prioritize Detection Effectiveness Reviews (DERs) across an enterprise MITRE ATT&CK-mapped detection portfolio by developing a Databricks forecasting dashboard in Python notebooks (Spark pipelines over blob-stored alert data) that flags detections nearing their review window and delivers monthly forecasts through a Slack bot.
- Closed coverage gaps across 8+ production detections by running DERs with synthetic data and live attack simulations against macOS and Windows test endpoints instrumented with CrowdStrike EDR.

Identity Security Engineer Intern, GE HealthCare – Waukesha, WI May 2025 – Aug 2025

- Cut privileged-access operational effort 25× and \$200K+ in annual licensing spend by automating CyberArk Privilege Cloud onboarding, safe provisioning, role assignments, and license management with REST APIs, Python, Bash, and PowerShell.
- Brought 2,600+ high-risk accounts (roughly 1,700 SOX and 970+ Ansible) under vaulted privileged-access control during an enterprise PAM rollout by onboarding them into CyberArk, strengthening least-privilege enforcement and compliance alignment.
- Accelerated team knowledge transfer and operational consistency by documenting identity-security processes in comprehensive technical documentation and standard operating procedures (SOPs).
- Kept identity services available to the business by administering core IAM platforms and resolving ServiceNow tickets for access provisioning, identity lifecycle management, and security incidents.

Technology Officer Volunteer, Sanskriti – Bolingbrook, IL Aug 2022 – Present

- Secured collaboration for a 50-member nonprofit by administering Google Workspace provisioning, access control, and security.
- Raised \$50K+ in donations over two years by developing and maintaining the organization's website, e-commerce platform, and integrations.

Information Technology Intern, Valley View School District – Bolingbrook, IL Aug 2022 – May 2023

- Reduced manual provisioning workload for a 3,600+ user district by automating device onboarding and account lifecycle across Google Admin Console and Okta LDAP, keeping access controls current.

CERTIFICATIONS

CompTIA Security+ Apr 2023

PROJECTS

Cloud Security & Networking Toolset , kapadia.org	May 2026 – Present
- Shipped 12 public cybersecurity and networking utilities on Cloudflare Pages, with GitHub Actions automating builds and deployments on every update to main. - Developed threat-intelligence tools that expose privacy risks in real-world web traffic, including an OSINT aggregator and a connection-analysis tool that evaluates browser fingerprinting, HTTP header exposure, and metadata leakage. - Secured serverless edge infrastructure with strict web-security policies, optimized API routing, and safe traffic handling while delivering DNS, subnetting, hashing, and token-decoding utilities.	
Homelab	Apr 2017 – Present
- Secured a full-stack home network by deploying OPNsense (DHCP, DNS, firewall, VPN, IDS/IPS) and Wazuh SIEM/XDR with custom detection rules, enforcing identity-aware access through Authentik IdP and Cloudflare Zero Trust. - Ran 22+ self-hosted services by managing Proxmox and Docker environments with reverse proxies, hosting containerized applications and smart devices via Home Assistant. - Practiced red/blue-team tradecraft with Kali, Nmap, Metasploit, and Burp Suite; performed forensic analysis with Autopsy, Volatility, and FTK Imager; and hosted redundant services across AWS, Azure, and GCP.	
GenAI Data Loss Prevention Platform , Capital One Intern Hackathon	Jul 2026
- Co-built, for an intern hackathon, a GenAI data-loss-prevention prototype spanning a Chrome extension, CLI, and Next.js web app that detected and redacted sensitive content in prompts and attachments sent to GenAI tools. - Developed regex and machine-learning classifiers on Microsoft Presidio to power real-time redaction across administrator-assignable Finance, Healthcare, and General PII domains.	
Responsible Vulnerability Disclosure	Oct 2025
Protected PII and academic records for 8,000+ students by identifying a critical access-control misconfiguration, documenting FERPA impact, and driving remediation through responsible disclosure to institutional leadership.	
CBORD Reverse-Engineered API	Oct 2024
Reverse-engineered the CBORD CS Access app's private HTTPS API with Burp Suite, then automated Siri-based door control through a Docker-deployed Python microservice and simplified REST API.	
Discord Bot	Sep 2023
Built a Python-based Discord bot integrated with the Proxmox REST API for secure, self-service VM lifecycle management, with real-time monitoring and automated idle-shutdown logic behind a permission-based portal.	

SKILLS

Security & Detection: Detection Engineering, Detection Effectiveness Reviews, MITRE ATT&CK, Sigma, YARA, SIEM/XDR (Splunk, Wazuh), CrowdStrike EDR (Detection Validation), Sysmon/Windows Event Logs, Malware Analysis, Vulnerability Analysis, Digital Forensics, OSINT	
Identity & Infrastructure: IAM, PAM (CyberArk Privilege Cloud), IGA (Saviynt), Authentik IdP, Okta LDAP, OAuth, Cloudflare Zero Trust, Least Privilege, ServiceNow, Ansible, MDM	
Cloud & Platforms: AWS, Azure, GCP, Cloudflare Pages, Serverless Computing, Databricks, Apache Spark, Blob Storage, Terraform (IaC), Docker, Proxmox, Slack, Git, GitHub Actions (CI/CD), OPNsense, Suricata, WireGuard, Linux, Windows, macOS	
Offensive & Forensics: ANY.RUN, Autopsy, Burp Suite, FTK Imager, Gobuster, Maltego, Metasploit, Nessus, Nmap, Volatility, Wireshark, Web Privacy Analysis, Browser Fingerprinting	
Development & Data: Python, Bash, PowerShell, Java, JavaScript, SQL, YAML, HTML/CSS, REST APIs, Next.js, Microsoft Presidio, AI-Assisted Development (Claude Code, GitHub Copilot), Prompt Engineering	
Concepts & Compliance: Containerization, Detection-Health Evaluation, Statistical Modeling, Generative AI Workflows, Infrastructure-as-Code, Virtualization, Zero Trust Architecture, SOX, HIPAA, PCI DSS, GDPR, FERPA, Cryptography, Safe API Design, OSI Model, TCP/IP	

LEADERSHIP & ACTIVITIES

Founder & Educator , Digital Literacy Program	Nov 2023 – Present
Brother Mentor , Sigma Phi Epsilon – Illinois Beta; former President, VP Recruitment, and VP Programming	Nov 2023 – Present
Scholar & Volunteer , Camras Scholars Program	Aug 2023 – Present